

MADHAVAN BALAJI

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TECHNICAL SKILLS

- **AI and LLM Frameworks:** LangGraph, CrewAI, LangChain, LlamaIndex, OpenAI APIs, HuggingFace Transformers, PEFT/LoRA, Stable Diffusion, Model Observability, Core Evaluation
- **Vector Search and RAG:** Qdrant, ChromaDB, FAISS, Sentence Transformers, Semantic Search, Retrieval-Augmented Generation (RAG), BM25, CrossEncoder Reranking
- **Deployment and Infrastructure:** FastAPI (Async), Docker, Kubernetes, AWS, GCP, WebSockets, RESTful APIs, CI/CD (GitHub Actions), Render Cloud, Git
- **Core ML and Vision:** PyTorch, TensorFlow, Scikit-learn, Vision Transformers (ViT), Text Classification, Entity Extraction, Python, SQL

EXPERIENCE

Technical Instructor & Curriculum Lead

Aug 2025 - Dec 2025

Indiana University Indianapolis

Indianapolis, IN

- Elevated academic performance across a **30+ student** cohort by architecting and delivering modular weekly workshops on core Data Structures, Algorithms, and optimization concepts through live-coding challenges.
- Boosted student engagement and benchmarking transparency by engineering a customized competitive programming leaderboard tracking active weekly and cumulative overall class rankings.
- Enhanced code optimization capabilities by coordinating structured peer-review labs evaluating execution efficiency, complex algorithmic edge cases, and Big-O space-time complexity trade-offs.

Data Analyst Intern

Sep 2022 - Nov 2022

Coincent

Bangalore, India

- Achieved **90% classification accuracy** and sub-second query times by engineering a **CNN-based** fashion retrieval pipeline translating unstructured image streams into a searchable index matrix.
- Cut model feature extraction latency by **30%** by profiling bottlenecks and implementing localized structural image preprocessing loops to unlock real-time streaming throughput.
- Generated a **40% lift** in simulated user engagement by developing a robust multi-class classifier spanning **1,000+ distinct clothing categories** to power downstream recommendation engines.

TECHNICAL PROJECTS

Sovereign SRE: Autonomous Self-Healing Infrastructure | [GitHub](#) | [Web App](#)

May 2026

- Designed a production **multi-agent** SRE system orchestrating a stateful **LangGraph** routing engine with a **CrewAI** collective for autonomous root-cause analysis and AI-generated patch generation, automating **100%** of initial production incident handling steps.
- Integrated a multi-model **LLM** layer connecting **Ollama (Llama 3.3)**, **Anthropic**, and **Gemini** behind programmatic **Human-in-the-Loop** validation gates, ensuring every AI-generated patch passed branch-verified **GitHub** Pull Request review before deployment.
- Established production AI observability by decoupling agent microservices via **Docker Compose** with asynchronous **WebSocket** telemetry, feeding live system state to a **Next.js** dashboard on **AWS Amplify** for real-time incident visibility.

DocuMind: Multi-Agent Research Intelligence Platform | [GitHub](#) | [Web App](#)

Jan 2026

- Architected a production **hybrid RAG** system achieving **167 texts/sec** throughput and **sub-15ms p50** retrieval latency by fusing dense **Qdrant** vector search, **BM25** sparse retrieval, and **CrossEncoder** reranking via Reciprocal Rank Fusion inside a stateful **LangGraph** orchestration workflow.
- Built a 4-agent **CrewAI** crew — Researcher, Analyst, Writer, Critic — with an iterative self-critique loop and open-sourced a custom **MCP server** registering 6 production tools for native **Claude Desktop** integration, eliminating workflow context-switching for AI-native users.
- Guaranteed zero data loss across concurrent LLM sessions by engineering a non-blocking **asyncpg/PostgreSQL** persistence layer with full **ACID** compliance, storing agent runs, retrieval audit trails, and session state backed by a background worker pool with live **WebSocket** telemetry.

DentaVision: Vision Transformer SaaS | [GitHub](#) | [Web App](#)

May 2025

- Deployed a production **Vision Transformer (ViT)** inference service achieving **92.4% top-1 accuracy** (+8.1 pts vs. CNN baseline), fine-tuned on **15K+** augmented intraoral dental images using **PyTorch** and **timm** with a deterministic Resize to ToTensor to Normalize preprocessing pipeline.
- Established a reproducible model evaluation framework by systematically benchmarking **5 deep learning architectures** against Accuracy, Precision, Recall, and F1-score across Caries, Gingivitis, and Tooth Discoloration disease classes.
- Optimized inference to **P95 latency of 310ms** via a **FastAPI** serving layer with single startup model loading, stateless request processing, and automatic **CUDA/CPU** device fallback, containerized in **Docker** and deployed to Render Cloud.

PATENT

Published: Caremate: An Integrated Elderly Support System | [View Document](#)

EDUCATION

Indiana University Indianapolis

May 2026

Master of Science in Computer Science

Indianapolis, IN

- **Relevant Coursework:** Deep Learning, Machine Learning, Computer Vision, Algorithms, Data Mining, Database Design, Natural Language Processing, Cloud Computing

Sri Venkateswara College of Engineering

Bachelor of Information Technology

Chennai, India